

## ASIC Outlook Wrap-up - MTK, MRVL, Intel Firming

Jeff Pu, CFA

SFC CE No. BNO719

jeffpu@gfgroup.com.hk

Henry Huang

SFC CE No. BYA749

henryhuang@gfgroup.com.hk

## What's New?

Following our recent earnings notes and supply chain research, we provide our view on major ASIC's volume and progress. We continue to like MRVL, MTK and Intel for the positive/intact outlook.

## Comments

**Nvidia vs. ASIC:** Contrary to general perception, we still believe that Nvidia's share will remain stable in 2027, on CoWoS consumption basis at TSMC; together with AMD, we expect GPU share is on an increasing trend (figure 1). As said in our recent *Nvidia Preview note on Aug 18*, we believe that Nvidia's Vera Rubin Platform is much ahead of its competitors in performance/TCO/supply chain and is gaining shares in major CSPs such as AWS/GOOG which even have their own ASICs.

**TPU dynamics—prefer MTK (2454 TT Buy):** We cut Broadcom's Sunfish volumes for 2026/27 on June 16 (our *report of June Edition*) due to packaging re-design, and set 460k CoWoS (ASIC) in 2027. Shipments of Sunfish should begin in late-2026 with volume in 1Q27. For 2027, we keep Broadcom's 50k CoWoS estimate to Meta for Iris+ and ARKE, but highlight the difficulty in 2H27 of reallocating capacity back to TPU as MTK's Humufish ramps on N2 advantage. On the other hand, we keep MTK's Zebrafish volumes unchanged at 25k/180k CoWoS for 2026/27. However, investor feedback suggests MTK is more like a funding-short stock amid concerns over Humufish tape-out, Google's slower AI model progress and worries over AMD/MRVL. However, we expect the engineering tape-out to be around Nov-2026, with Google/MTK committing meaningful resources and a manageable pass-rate bar. We continue to prefer MTK over Broadcom.

**Google attach silicon—prefer MRVL (Buy):** As said in our *MRVL report on Aug 20*, we continue to see meaningful contribution from Google's silicon—CXL, DPU, and MPU. We expect ~\$3bn each from CXL and DPU by 2028. For MPU, optimized for token generation, we model a 2Q28 start on TSMC N2 and conservatively estimate ~\$3bn CY28 revenue. We also view Google's equity partnership as increasing MRVL's chances of securing TPU v10 (Meropé) in 2H28. This won't impact MTK, in our view, given that the Icefish engagement already confirmed.

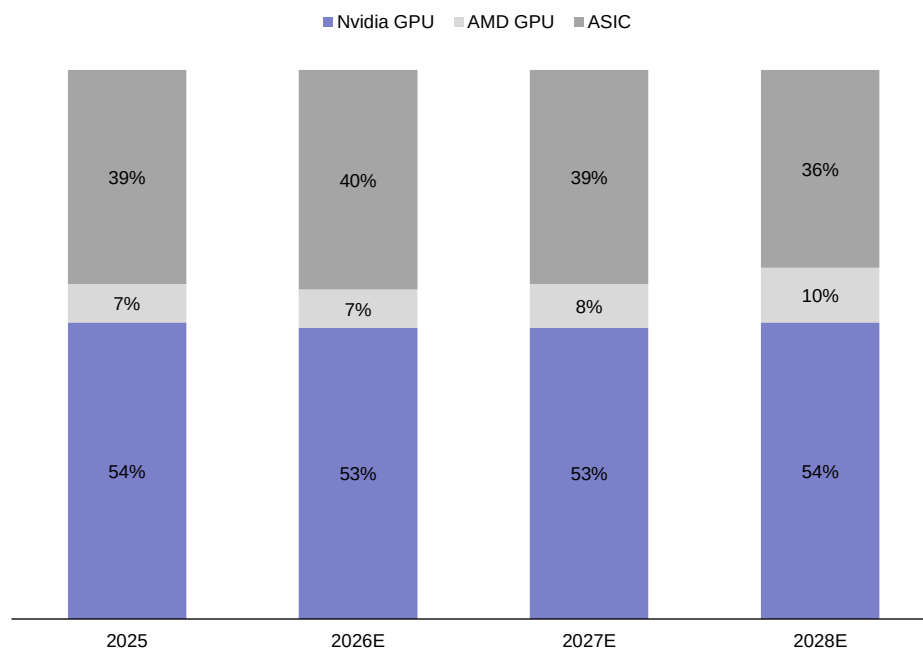
**AWS T3/T4—prefer MRVL (Buy) and SMT (Buy):** We continue to expect Marvell to participate in Trainium 3, contributing roughly \$1bn/\$2bn in CY26/27. For Trainium 4, given the design of NPO scale-up for UALink/NVLink, we believe that Marvell's strengthened scale-up optical networking position post-Celestial AI will lead to an enhanced allocation share, vs. its competitors. Together with a much higher ASP, we expect >\$3bn revenue contribution of T4 to MRVL in CY28. For NPO, we expect 6.4T OE to be the primary specs for T4. We forecast AWS will consume 5m/12m OE units for Trainium 4 in 2H27E/2028E. We forecast global OE units to be 11m/40m at optical component's level and like SMT (Buy) for its 2<sup>nd</sup> supplier position.

**ASIC packaging—prefer Intel (Buy) and ABF substrate names:** Per our *Intel note on Aug 14*, we raised our back-end revenue forecast to \$1.1bn/\$7bn in FY27/FY28, supported by better visibility on AWS T3 for EMIB-T in 2027 and major Google Humufish/Triggerfish volumes through 2H27–2028. Looking ahead, we see a path with high likelihood to win back-end business from U.S. fabless including MSFT, QCOM, MRVL, with products across ASIC, CPU and CPO. On supply, we expect ABF substrate to greatly benefit from EMIB.

**Risks:** 1) AI demand slowdown, 2) infrastructure deployment, 3) supply constraints, 4) competition.

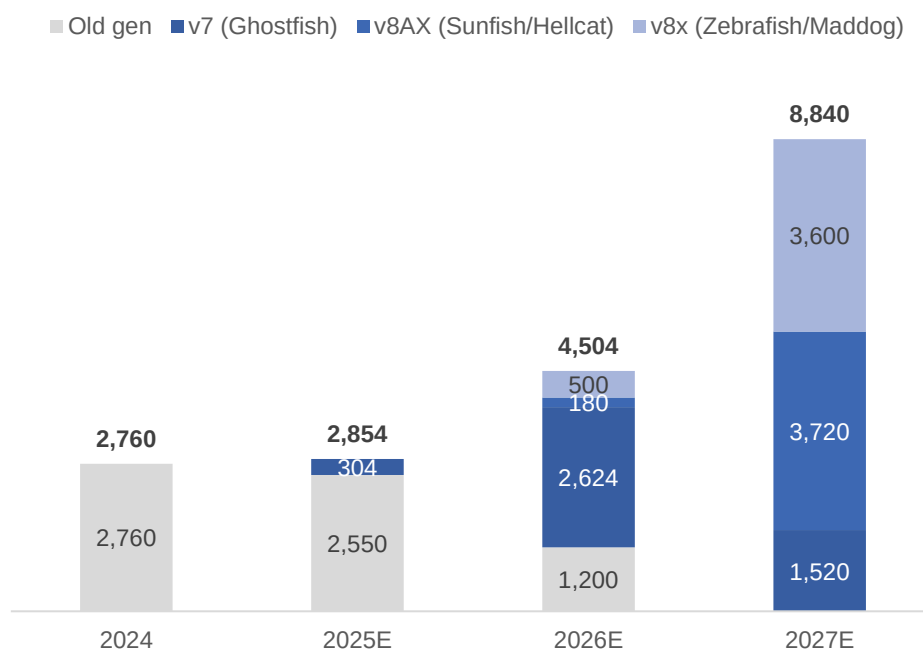
## Focus Charts

Figure 1: GPU, vs. ASIC's CoWoS share at TSMC



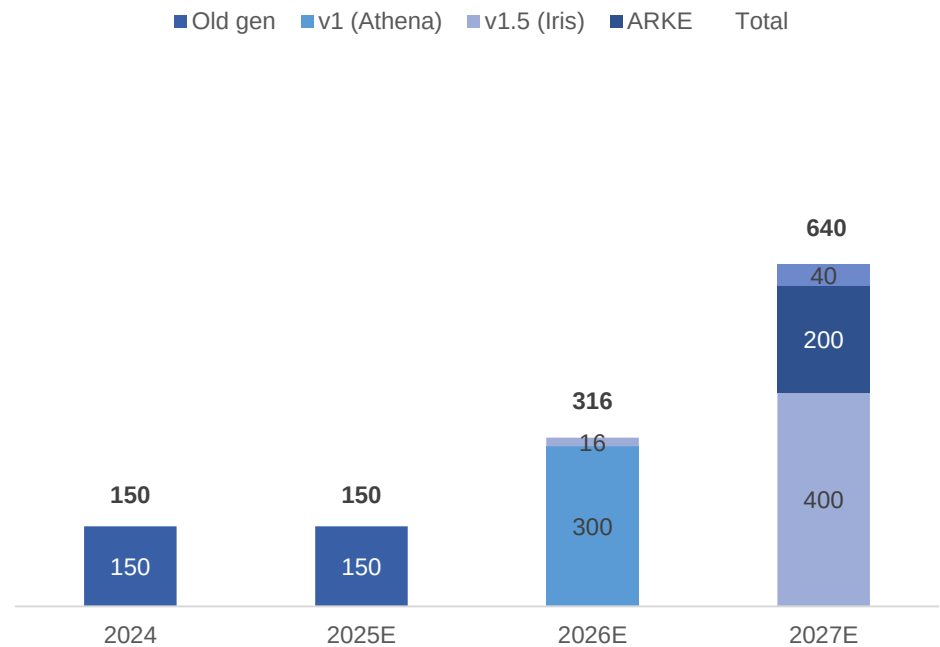
Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 2: Google TPU volume estimates (units mn)



Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 3: Meta ASIC volume estimates (units mn)



Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 4: Google TPU Roadmap

|                              | TPU v7<br>Ironwood | TPU 8i           | TPU 8t           | TPU v9t            | TPU v9t+            | TPU v9i             | TPU v10t             |
|------------------------------|--------------------|------------------|------------------|--------------------|---------------------|---------------------|----------------------|
| Code name                    | Ghostfish          | Sunfish          | Zebrafish        | Humufish           | Triggerfish         | Whalefish           | Icefish              |
| Time                         | 4Q25               | 4Q26             | 4Q26             | 4Q27               | 4Q27                | 1Q28                | 4Q28                 |
| Design Service -<br>Compute  | BRCM               | BRCM             | GOOG             | GOOG               | GOOG                | BRCM                | GOOG                 |
| Design Service -<br>I/O      | BRCM               | BRCM             | MTK              | MTK                | MTK                 | BRCM                | MTK                  |
| Design Service -<br>back-end | BRCM               | BRCM             | MTK              | MTK                | MTK                 | BRCM                | GOOG or MTK          |
| Compute                      | 2*N3E              | 2*N3P            | 1*N3P            | 4*N2P              | 4*N2P               | 2*N3P               | A14/N2P*8+<br>N2P*16 |
| I/O                          | 1*N5               | 1*N4             | 1*N4P            | 4*N3P              | 4*N3P               | 2*N4                | N2P*6                |
| Memory                       | 8*HBM3e<br>8-Hi    | 8*HBM3e<br>12-Hi | 6*HBM3e<br>12-Hi | (6*HBM4 8Hi)<br>*2 | (6*HBM4e 8Hi)<br>*2 | (6*HBM4e 8Hi)<br>*2 | 16*HBM5<br>12-Hi     |
| Memory spec                  | 192GB              | 288GB            | 216GB            | 288GB              | 384GB               | 384GB               |                      |
| Memory BW<br>(TB/s)          | 7.4                | 8.6              | 6.5              |                    |                     |                     |                      |
| Back-end                     | CoWoS-S            | CoWoS-L          | CoWoS-S          | EMIB-T             | EMIB-T              | CoWoS-S/SOIC        | EMIB-T or<br>CoWoS   |

Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 5: Meta ASIC Roadmap

|                           | MTIA 300     | MTIA 400     | MTIA 400+      | MTIA 450       | MTIA 500    |
|---------------------------|--------------|--------------|----------------|----------------|-------------|
| Code name                 | Athena       | Iris Plus    | Iris Plus+     | ARKE           | Astrid      |
| Time                      | 2Q26         | 2Q26         | 3Q26           | 1Q27           | 3Q27        |
| Design Service - compute  | BRCM         | BRCM         | BRCM           | BRCM           | BRCM        |
| Design Service - I/O      | BRCM         | BRCM         | BRCM           | BRCM           | BRCM        |
| Design Service - back-end | BRCM         | BRCM         | BRCM           | BRCM           | BRCM        |
| Compute                   | 1*N3E        | 1*N3         | 2*N3           | 1*N2           | N2          |
| I/O                       | 2*N4         | 2*N4         | 2*N3 +1 N3 SoC | 2*N3 +1 N3 SoC | N3          |
| Memory                    | 6*HBM3e 12Hi | 6*HBM3E 12Hi | 8*HBM3E 12Hi   | 8*HBM4 12Hi    | 8*HBM4 12Hi |
| Memory spec               | 216GB        | 216GB        | 288GB          | 288GB          | 384GB       |
| Memory BW (TB/s)          | 6.1          | 9.2          | 9.2            | 12.8           | TBD         |
| TDP (W)                   | 800          | 1000         | 1200           | 1400           | TBD         |
| Back-end                  | CoWoS-S      | CoWoS-L      | CoWoS-L        | CoWoS-L        | CoWoS-L     |

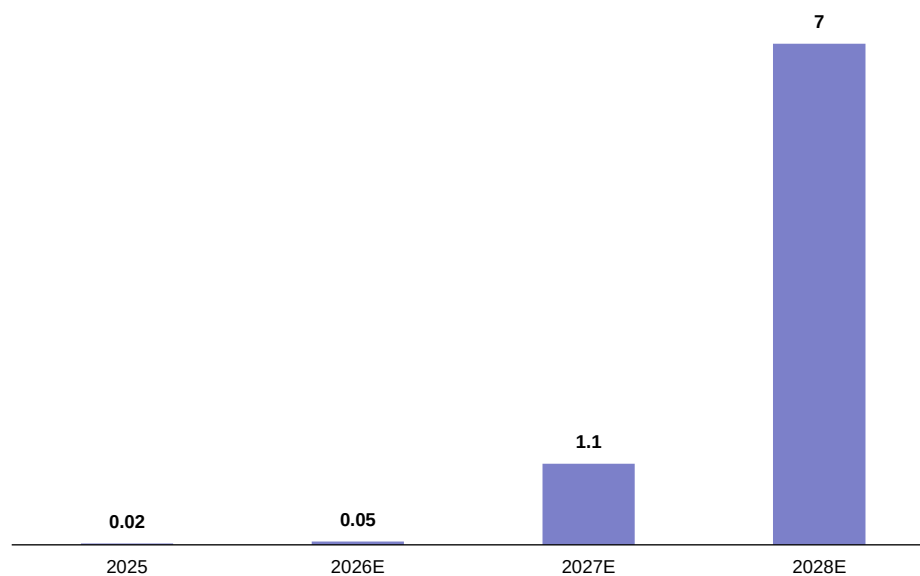
Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 6: NPO OE shipment forecast

|                                                        | 2027       | 2028        |
|--------------------------------------------------------|------------|-------------|
| <b>Market for Nvidia's scale up 3.2T NPO (k units)</b> |            |             |
| Rubin Ultra chip for NVL576(m units)                   | 1.0        | 6           |
| NPO Penetration rate                                   | 60%        | 95%         |
| OE per GPU/switch ASIC                                 | 4          | 4           |
| <b>OE</b>                                              | <b>2.3</b> | <b>22.8</b> |
| Considering pull-in                                    | 6          | 19          |
| <b>Market for AWS's scale up 6.4T NPO (k units)</b>    |            |             |
| Trainium 4 in total (m units)                          | 0.8        | 4.5         |
| NPO Penetration rate                                   | 80%        | 80%         |
| Trainium 4 for Scale Up (m units)                      | 0.6        | 3.6         |
| OE per GPU/switch ASIC                                 | 4          | 4           |
| <b>OE</b>                                              | <b>2.6</b> | <b>14.4</b> |
| Considering pull-in                                    | 5          | 12          |
| <b>Market for Other CSPs' scale up NPO (k units)</b>   |            |             |
| <b>OE</b>                                              |            | <b>9.0</b>  |
| <b>Market for Total scale up NPO (k units)</b>         |            |             |
| <b>Total</b>                                           | <b>4.8</b> | <b>46.2</b> |
| Considering pull-in                                    | 11         | 40          |

Sources: Company data, GF Securities (Hong Kong) Brokerage

**Figure 7: Intel's backend revenue forecast (USD bn)**



Sources: Company data, GF Securities (Hong Kong) Brokerage

### Risks

1) AI demand slowdown, 2) infrastructure deployment, 3) supply constraints, 4) competition.

## Rating definitions Benchmark: Hang Seng Index (Hong Kong)

|                        |                     |                                                                 |
|------------------------|---------------------|-----------------------------------------------------------------|
| <b>Company ratings</b> | <b>Buy</b>          | Stock expected to outperform benchmark by more than 10%         |
|                        | <b>Hold</b>         | Expected stock relative performance ranges between -10% and 10% |
|                        | <b>Underperform</b> | Stock expected to underperform benchmark by more than 10%       |

|                       |                 |                                                                  |
|-----------------------|-----------------|------------------------------------------------------------------|
| <b>Sector ratings</b> | <b>Positive</b> | Sector expected to outperform benchmark by more than 10%         |
|                       | <b>Neutral</b>  | Expected sector relative performance ranges between -10% and 10% |
|                       | <b>Cautious</b> | Sector expected to underperform benchmark by more than 10%       |

|                          |                                                       |
|--------------------------|-------------------------------------------------------|
| <b>Hong Kong Company</b> | GF Securities (Hong Kong) Brokerage Limited           |
| <b>Address</b>           | 27/F, GF Tower, 81 Lockhart Road, Wan Chai, Hong Kong |
| <b>Telephone</b>         | (852) 37191111                                        |
| <b>Email</b>             | evanlee@gfgroup.com.hk                                |

## Legal Disclaimer

This report has been prepared by GF Securities (Hong Kong) Brokerage Limited ("GF Securities (Hong Kong) Brokerage"). According to the laws, regulations and regulatory requirements in different countries and regions, this report is distributed by GF Securities (Hong Kong) Brokerage with relevant legal and compliant operation qualifications in these countries and regions.

GF Securities (Hong Kong) Brokerage is licensed by the Securities and Futures Commission of Hong Kong ("SFC") to conduct Type 4 Regulated Activity "Advising on Securities". It is regulated by the SFC, and is responsible for the distribution of this report in Hong Kong. Information about the qualifications of the research analyst(s) who is(are) the author(s) of this report as licensed by the SFC are disclosed in the section where research analyst names are shown.

The research analyst(s) primarily responsible for the content of this report, in whole or in part, certifies that with respect to the company or relevant securities that the analyst(s) covered in this report: 1) all of the views expressed accurately reflect his or her personal views on the company or relevant securities mentioned herein; and 2) no part of his or her remuneration was, is, or will be, directly or indirectly, in connection with his or her specific recommendations or views expressed in this report.

This report is published solely for information purpose and does not constitute an offer to buy or sell any securities or a solicitation of an offer to buy, or recommendation for investment in, any securities.

The securities mentioned in this report may not be allowed to be sold in certain jurisdictions. No action has been taken to permit the distribution of the research report to any person in any jurisdiction that the circulation or distribution of such research report is unlawful. This report is distributed solely to clients or designated institutions authorized by GF Securities (Hong Kong) Brokerage, and is not distributed publicly. It is distributed to certain clients based on the conclusion that they are able to assess investment risks independently, execute investment decisions independently and assume corresponding risks independently.

This report has been issued and based on information obtained from sources generally available to the public and believed by the research analyst(s) to be reliable but which has not been independently verified. No representation or warranty, either express or implied, is made by GF Securities (Hong Kong) Brokerage as to their accuracy and completeness of the information contained in this report. GF Securities (Hong Kong) Brokerage accepts no liability for all loss arising from the use of the materials presented in this report, unless is excluded by applicable laws or regulations. Please be aware of the fact that investments involve risks and the price of securities may be fluctuated and therefore return may be varied, past results do not guarantee future performance. Any recommendation contained in this report does not have regard to the specific investment objectives, financial situation and the particular needs of any individuals. This report is not to be taken in substitution for the exercise of judgment by respective recipients of this report, where necessary, recipients should obtain professional advice before making investment decisions.

GF Securities (Hong Kong) Brokerage may have issued, and may in the future issue, other communications that are inconsistent with, and reach different conclusions from, the information presented in the research report. The points of view, opinions and analytical methods adopted in the research report are solely expressed by the analysts but not that of GF Securities (Hong Kong) Brokerage or its affiliates. The information, opinions and forecasts presented in the research report are the current opinions of the analysts as of the date appearing on this material only which may subject to change at any time without notice. The salesperson, dealer or other professionals of GF Securities (Hong Kong) Brokerage may deliver opposite points of view to their clients and the proprietary trading division with respect to market commentary or dealing strategy either in writing or verbally. The proprietary trading division of GF Securities (Hong Kong) Brokerage may have different investment decision which may be contrary to the opinions expressed in the research report. GF Securities (Hong Kong) Brokerage or its affiliates or respective directors, officers, analysts and employees related to research report business may have rights and interests in securities mentioned in the research report. Recipients should be aware of relevant disclosure of interest (if any) when reading this report.

GF Securities (Hong Kong) Brokerage and its affiliates may be seeking or building business relationships with company(ies) mentioned in this report. Therefore, investors should consider the impact on the independence of this report by GF Securities (Hong Kong) Brokerage and its affiliates due to potential conflicts of interests. Investors should not make any investment decisions based solely on the contents of this report. Investors should make their own investment decisions and bear their own risk. No written or verbal commitment of sharing gains or losses from securities investments in any form shall be effective.

This report may contain and/or describe/present factual historical information on prices of Futures contracts (the "information"). Please note that this information is solely for the purpose of forming part of the argument/grounds/evidence in our research methodology/analysis to support our conclusion on our view of the relevant industry/company mentioned. It does not, by any means (express or implied) to be associated with or constituted as SFC Type 5 Regulated Activity (Advising on futures contracts).

## Important Notices

GF Securities Co. Ltd and its affiliates may be seeking or building business relationships with companies mentioned in this report. Therefore, investors should consider the impact on the independence of this report by GF Securities Co. Ltd and its affiliates due to potential conflicts of interests. Investors should not make any investment decisions based solely on the contents of this report.

The research analyst(s) primarily responsible for the content of this research report, in whole or in part, certifies that with respect to the company or relevant securities that the analyst(s) covered in this report: (1) all of the views expressed accurately reflect his or her personal views on the company or relevant securities mentioned herein; and (2) no part of his or her remuneration was, is, or will be, directly or indirectly, in connection with his or her specific recommendations or views expressed in this research report.

The remuneration for the production of this report by research staff is determined based on various factors including the quality of research, client review and workload, and it is also affected by factors including GF Securities' overall revenue, part of which is generated from GF Securities' investment banking business.

This report is distributed solely to clients or designated institutions authorized by GF Securities, and is not distributed publicly. It can only be used by the targeted recipients who have the obligation to maintain confidentiality. GF Securities does not view any persons who have received or read this report through other means as its clients.

This report is distributed to a certain client based on the conclusion that this client is deemed to be able to assess investment risks independently, execute investment decisions independently and assume corresponding risks independently.

The contents of this report is for reference only. Information contained or opinions expressed in this report does not constitute an offer or inquiry. Customers should not substitute their independent judgment with this report or make decisions based on this report.

This report is published solely for information purpose and does not constitute an offer to buy or sell any securities or a solicitation of an offer to buy, or recommendation for investment in, any securities. The research report is intended solely for use of the clients of GF Securities. The securities mentioned in the research report may not be allowed to be sold in certain jurisdictions. No action has been taken to permit the distribution of the research reports to any person in any jurisdiction that the circulation or distribution of such research report is unlawful. No representation or warranty, either express or implied, is made by GF Securities as to their accuracy and completeness of the information contained in the research report. GF Securities accepts no liability for all loss arising from the use of the materials presented in the research report, unless is excluded by applicable laws or regulations. Please be aware of the fact that investments involve risks and the price of securities may be fluctuated and therefore return may be varied, past results do not guarantee future performance. Any recommendation contained in the research report does not have regard to the specific investment objectives, financial situation and the particular needs of any individuals. The report is not to be taken in substitution for the exercise of judgment by respective recipients of the report, where necessary, recipients should obtain professional advice before making investment decisions.

GF Securities may have issued, and may in the future issue, other communications that are inconsistent with, and reach different conclusions from, the information presented in the research report. The points of view, opinions and analytical methods adopted in the research report are solely expressed by the analysts but not that of GF Securities or its affiliates. The information, opinions and forecasts presented in the research report are the current opinions of the analysts as of the date appearing on this material only which may subject to change at any time without notice. The salesperson, dealer or other professionals of GF Securities may deliver opposite points of view to their clients and the proprietary trading division with respect to market commentary or dealing strategy either in writing or verbally. The proprietary trading division of GF Securities may have different investment decision which may be contrary to the opinions expressed in the research report. GF Securities or its affiliates or respective directors, officers, analysts and employees related to research report business may have rights and interests in securities mentioned in the research report. Recipients should be aware of relevant disclosure of interest (if any) when reading the report.

## Disclosure of Interests

- 1) The research analyst(s) and his/her associate has not served as an officer of the company(ies) mentioned in this report, and does not have any financial interests in the company(ies) mentioned in this report.
- 2) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any financial interests and has not invested in interests aggregate to an amount equal to or more than (i) 1% of the market capitalization; or (ii) 1% of the issued share capital, or issued units, in the company(ies) mentioned in this report.
- 3) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any market making activities, and has not employed any individual(s) serving as officer(s) of the company(ies) mentioned in this report.
- 4) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any investment banking relationship in Hong Kong with the company(ies) mentioned in this report in the past 12 months.

Copyright © GF Securities (Hong Kong) Brokerage

Without the prior written consent obtained from GF Securities (Hong Kong) Brokerage, any part of the materials contained herein should not (i) in any forms be copied or reproduced or (ii) be re-disseminated.